ALGEBRIC REPRESENTATION

- SOP Sum of Product
- POS Product of sum

SOP REPRESENTATION

- A, B, C Literal
- AB, AC, BC, $\bar{A}B$, $A\bar{C}$, $\bar{B}\bar{C}$ — Product Term
- ABC, $\bar{A}BC$, $A\bar{B}C$, ... $\bar{A}\bar{B}\bar{C}$

Minterm : Girişlerin hepsini içerdiği zaman

Cannonical Product Term
Fundamental Product Term

→ Minterm = cannonical Product Term = Fundamental P. T.

$f(A,B,C) = A + \bar{B}C + \bar{A}BC$
↳ SOP Form

MINTERM CODE

Uncomplemented variable x logic 1

Complemented variable $\bar{x}$ logic 0

$m_i = \bar{A}BC = m_3$

(011) → 3   i=3

* cannonical SOP Form → Eğer çarpım terimlerinin tümü minterm ise buna denir.

ÖRN $f(A,B,C) = \bar{A}B\bar{C} + AB\bar{C} + \bar{A}BC + ABC$

cannonical SOP form

$(010) + (110) + (011) + (111)$
$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
$2 \quad 6 \quad 3 \quad 7$

Numerical Representation $= m_2 + m_3 + m_6 + m_7$

$= \Sigma m(2,3,6,7)$

logik fonksiyonun numerik gösterimi

→ SOP form fonksiyonun lojik 1 değerini aldığı noktalara ilişkin bir gösterimdir.

Minterm	Dec	A	B	C	f
m_0	0	0	0	0	0
m_1	1	0	0	1	0
m_2	2	0	1	0	1
m_3	3	0	1	1	1
m_4	4	1	0	0	0
m_5	5	1	0	1	0
m_6	6	1	1	0	1
m_7	7	1	1	1	1

$f = \Sigma m(2,3,6,7)$

$m_2 + m_3 + m_6 + m_7$

$f(A,B,C) = \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

→ POS form fonksiyonun lojic 0 değerini aldığı noktalara ilişkin bir gösterimdir.

For Generating the Cannonical SOP Form

$f(A,B,C) = A + \bar{B}C + \bar{A}BC$

1) $x + \bar{x} = 1$

$f(A,B,C) = A(B+\bar{B}).(C+\bar{C}) + (A+\bar{A}).B.C + \bar{A}BC$

2) Distributive

$f(A,B,C) = ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}BC$

3) $x + x = x$ → idempotand

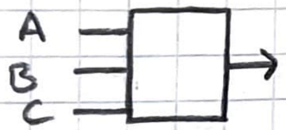
$f = ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$

$f = \Sigma m(7,6,5,4,1,3)$

$f = \Sigma m(1,3,4,5,6,7)$

Dec	A	B	C	f
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

POS REPRESENTATION (Product Of Sum)



- $A, B, C, \bar{A}, \ldots, \bar{C}$ Literal
- $A+B,\ A+C,\ B+C, \ldots\ \bar{B}+\bar{C}$
- $A+B+C,\ A+\bar{B}+\bar{C}, \ldots\ \bar{A}+\bar{B}+\bar{C}$) Maxterm

MAXTERM CODE (M_i)

$$\left.\begin{array}{l}\text{Completed variable } \bar{x} \text{ logic } 1 \\ \text{Uncompleted variable } x \text{ logic } 0\end{array}\right\} \quad \bar{A}+\bar{B}+C+\bar{D} = M_{13}$$

$(1\ 1\ 0\ 1)_2 \rightarrow 13$

$\rightarrow \overline{M_i} = m_i$

ÖRN $m_5 = A\bar{B}C$

$$\overline{m_5} = \overline{A\bar{B}C} = \bar{A}+B+\bar{C} \Rightarrow M_5$$

$(1\ 0\ 1)$

$$F(A,B,C) = (A+B+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+B+C)(A+B+C)$$

$(0\ 0\ 1)$ $(1\ 0\ 1)$ $(1\ 0\ 0)$ $(0\ 0\ 0)$

M_1 M_5 M_4 M_0

$$= M_0 \cdot M_1 \cdot M_4 \cdot M_5$$

$$= \prod M(0,1,4,5)$$

Dec	A	B	C	F
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

1) From SOP to POS Form

$$f(x,y,z) = x\bar{z} + \bar{x}y + \bar{y}z$$

De Morgan Teoremi

$$f^{d}(x,y,z) = (\bar{x}+z)\cdot(x+\bar{y})\cdot(y+\bar{z})$$

$$f^{d}(x,y,z) = \bar{x}\bar{y}\bar{z} + xyz$$

$$f^{dd}(x,y,z) = f(x,y,z) = \underset{0\,0\,0}{(x+y+z)}\cdot\underset{1\,1\,1}{(\bar{x}+\bar{y}+\bar{z})} = (0,7)$$

$$\underbrace{(\bar{x}\cdot x\cdot y)}_{0} + \underbrace{(\bar{x}\cdot x\cdot \bar{z})}_{0} + \underbrace{(\bar{x}\cdot\bar{y}\cdot y)}_{0} + (\bar{x}\cdot\bar{y}\cdot\bar{z}) + (z\cdot x\cdot y) +$$

$$\underbrace{(z\cdot x\cdot\bar{z})}_{0} + \underbrace{(z\cdot\bar{y}\cdot y)}_{0} + \underbrace{(z\cdot\bar{y}\cdot\bar{z})}_{0}$$

2) From POS to SOP Form

$$f(x,y,z) = (x+y)\cdot(\bar{x}\cdot z) = x\cdot\bar{x} + x\cdot z + \bar{x}\cdot y + y\cdot z$$

$$f_{SOP} = xz + \bar{x}y + yz$$

KARNAUGH MAP

1)

Dec	x	$F=\bar{x}$
0	0	1
1	1	0

x	f
0	1
1	0

f_{SOP}

$$\underbrace{f_{SOP} = \bar{x} \qquad f_{POS} = \bar{x}}_{\text{Algebra}} \qquad \underbrace{f_{SOP} = \Sigma m(0) \qquad f_{POS} = \Pi M(1)}_{\text{Numeric}}$$

2)

Dec	A	B	F
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

A \ B	0	1
0	1_0	0_1
1	0_2	1_3

f

B \ A	0	1
0	1_0	0_2
1	0_1	1_3

f

$$f_{SOP} = \bar{A}\bar{B} + AB \quad (0,3)$$

$$f_{POS} = (A+\bar{B}) \cdot (\bar{A}+B) \quad (1,2)$$

3)

Dec	A	B	C	f
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

← Gray code

A \ BC	00	01	11	10
0	0_0	1_1	0_3	1_2
1	0_4	1_5	0_7	1_6

f

$$f_{SOP} = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$$

$$f_{POS} = (A+B+C) \cdot (A+\bar{B}+\bar{C}) \cdot (\bar{A}+B+C) \cdot (\bar{A}+\bar{B}+\bar{C})$$

$$f_{SOP} = \Sigma m(1,2,5,6)$$

$$f_{POS} = \Pi M(0,3,4,7)$$

AB \ CD	0	1
00	0_0	1_1
01	1_2	0_3
11	1_6	0_7
10	0_4	1_5

4)

Dec	A	B	C	D	f
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

Gray code

AB \ CD	00	01	11	10
00	0 (0)	1 (1)	0 (3)	1 (2)
01	0 (4)	0 (5)	1 (7)	1 (6)
11	1 (12)	1 (13)	1 (15)	0 (16)
10	1 (8)	0 (9)	0 (11)	0 (10)

f

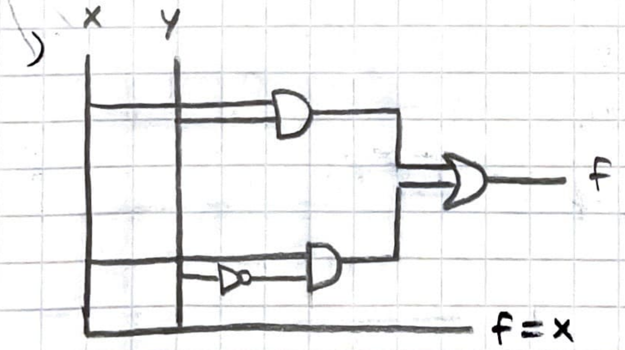
Gray code

LOJİK KOMŞULUKLARI

$A\bar{B}C + \bar{A}\bar{B}C$ } 2 and 1 or 2 inverter / 11 inputs

→ $xy + x\bar{y} = x(y+\bar{y}) = x.1 = x$

$A.x + \bar{A}.x = (A+\bar{A}).x = 1.x = x = \bar{B}C$ } 1 and 1 inverter / 3 inputs



2 and
1 or
1 inverter + 7 input

$\begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$

LOJİK ADJACENT

A \ B	0	1
0	1	1
1	0	1

$\bar{A}$

Symmetry Axes

B

f

$f_{SOP} = \bar{A} + B$

$f_{POS} = \bar{A} + B$

$f_{SOP} = \bar{A}\bar{B} + \bar{A}B + AB$

$= \bar{A}\bar{B} + B(\cancel{\bar{A}+A})^{1}$

$= \bar{A}\bar{B} + B = (\bar{A}+B)\cdot(\cancel{B+\bar{B}})^{1}$

$= \bar{A}+B$

A \ BC	00	01	11	10
0	1	0	0	1
1	1	1	1	0

$A+\bar{C}$

$\bar{A}\bar{C}$

$\bar{B}\bar{C}$

AC

$\bar{A}+\bar{B}+C$

$f_{SOP} = \bar{A}\bar{C} + \bar{B}\bar{C} + AC$

$P_{POS} = (A+\bar{C})\cdot(\bar{A}+\bar{B}+C)$

ÖRN

A \ BC	00	01	11	10
0	1	1	1	1
1	1	0	0	1

f

$\bar{A}$

$\bar{C}$

$\bar{A}+\bar{C}$

$f_{SOP} = \bar{A}\cdot\bar{C}$ (min maliyet)

$f_{POS} = \bar{A}+\bar{C}$

ÖRN

A \ BC	00	01	11	10
0	1	1	1	1
1	1	1	1	1

f

$f = 1$

ÖRN

AB \ CD	00	01	11	10
00	1 (0)	(1)	(3)	1 (2)
01	(4)	1 (5)	1 (7)	(6)
11	(13)	(14)	(16)	1 (15)
10	(8)	(9)	(12)	1 (11)

f

$\bar{A}\bar{B}\bar{D}$

$\bar{A}.B.D$

$A.C\bar{D}$

$f_{sop} = \bar{A}\bar{B}\bar{D} + \bar{A}BD + AC\bar{D}$

ÖRN

AB \ CD	00	01	11	10
00	1	1	1	1
01		1	1	1
11				1
10	1			1

f

$C\bar{D}$

$\bar{B}\bar{D}$ $\bar{A}D$

$f_{sop} = \bar{A}D + C\bar{D} + \bar{B}\bar{D}$

ÖRN

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11			1	1
10			1	1

f

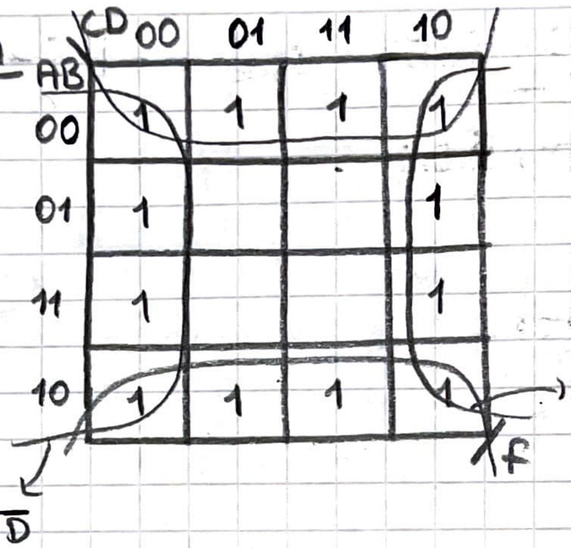
$\bar{A}$

$\bar{A}C$

$f_{sop} = \bar{A} + C$

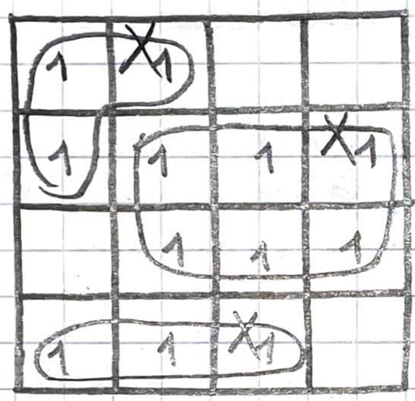
$\Rightarrow$ $2^n = 2^3 = 8$'li gruplar en yüksek sayıda old. için en düşük maliyetlidir. Bu yüzden 8li grup oluşturduk.

ÖRN



$f_{SOP} = \bar{B} + \bar{D}$

ÖRN



Bu komşuluklar mümkün değildir !

Çünkü 2^n'li komşuluklar oluşturmalıyız.

$2^1 \rightarrow$ 2'li grup
$2^2 \rightarrow$ 4'lü grup
$2^3 \rightarrow$ 8'li grup

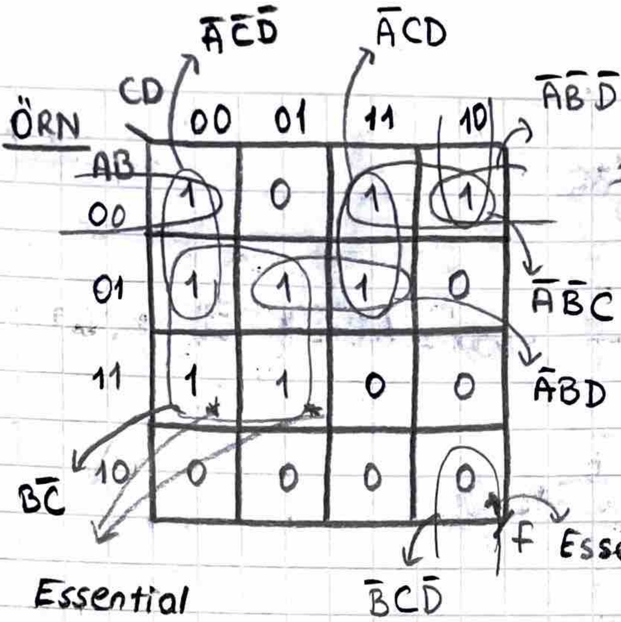
\# Prime Implicant (PI) (Temel İçeren) (Hepsi yazılır)

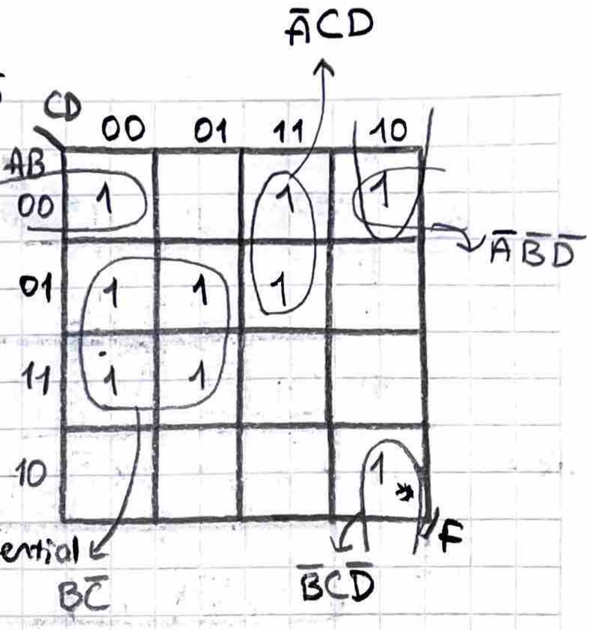
\# Essential Prime Implicant (EPI) (Gerekli temel içeren) (zorunlu olanlar)

\# Optional Prime Implicant (OPI) (seçimli) (isteğe bağlı olanlar)

\# Redundant Prime Implicant (RPI) (Gereksizler)

ÖRN





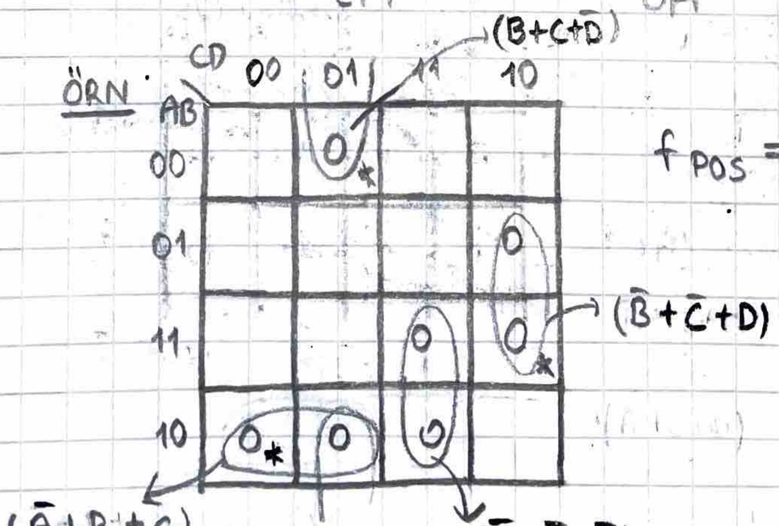
Gerekli Temel İçeren

PI	EPI	OPI	RPI
$B\bar{C}$	★ $B\bar{C}$	$\bar{A}\bar{B}\bar{D}$ ★	$\bar{A}\bar{C}\bar{D}$
$\bar{A}\bar{C}\bar{D}$	★ $\bar{B}C\bar{D}$	$\bar{A}CD$	$\bar{A}BD$
$\bar{A}\bar{B}\bar{D}$			$\bar{A}\bar{B}C$
$\bar{A}BD$			
$\bar{A}CD$			
$\bar{A}\bar{B}C$			
$\bar{B}C\bar{D}$			

→ Karnaugh haritasında indirgeme yaparken önce zorunlu olanlardan başla. (EPI).

$$f_{SOP} = \underbrace{B\bar{C} + \bar{B}C\bar{D}}_{EPI} + \underbrace{\bar{A}CD + \bar{A}\bar{B}\bar{D}}_{OPI} \rightarrow \bar{A}\bar{C}\bar{D}$$

ÖRN



$$f_{POS} = (\bar{A}+B+C).(\bar{A}+\bar{C}+\bar{D}).(\bar{B}+\bar{C}+D).(B+C+\bar{D})$$

→ indirgemeye başladığında önce zorunluları belirle, yıldız koy.

ÖRN

AB \ CD	00	01	11	10
00			1*	
01	1*	1	1	
11		1	1	1*
10		1*		

$\bar{A}BC$ · ABC · $\bar{A}B\bar{C}$ · $A\bar{C}D$ · BD

★ Güzel örnek dedi

$f_{SOP} = \bar{A}B\bar{C} + A\bar{C}D + ABC + BD + \bar{A}BD$

Incompletely Specified Functions (DON'T CARES)

$f(A,B,C) = \Sigma m(0,1,5,7) + \phi(2,4)$

$g(A,B,C,D) = \Pi M(0,1,4,6,8,14,15) \cdot \phi(2,3,9)$

A \ BC	00	01	11	10
0	1	1	0	ϕ
1	ϕ	1	1	0

f

$\bar{B}$ · AC

$f_{SOP} = \bar{B} + AC$

A \ BC	00	01	11	10
0	1	1	0	ϕ
1	ϕ	1	1	0

F

$(A+\bar{B})$ · $(\bar{B}+C)$ · $(\bar{A}+C)$

$f_{POS} = (A+\bar{B}) \cdot (\bar{A}+C)$ veya $(A+\bar{B}) \cdot (\bar{B}+C)$

AB \ CD	00	01	11	10
00	0	0	X	X
01	0	1	1	0
11	1	1	0	0
10	0	X	1	1

g

$(A+D)$ · $(\bar{A}+\bar{B}+\bar{C})$ · $(B+C)$

AB \ CD	00	01	11	10
00	0	0	X	X
01	0	1	1	0
11	1*	1	0	0
10	0	X	1	1

g

$\bar{A}BD$ · $AB\bar{C}$ · $\bar{B}C$

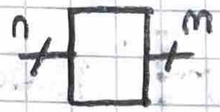
$g_{POS} = (B+C) \cdot (A+D) \cdot (\bar{A}+\bar{B}+\bar{C})$

$g_{SOP} = \bar{B}C + AB\bar{C} + \bar{A}BD$

→ Dont Care : 0 da 1 de olabilir. işime indirgemede hangisi yararsa onu alıyorum.

$\emptyset$, x

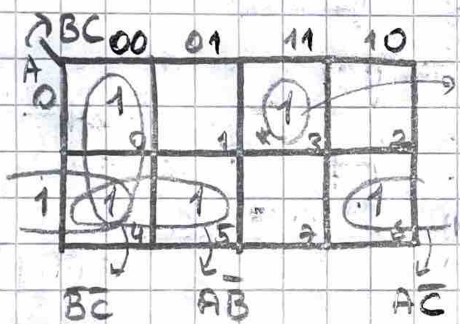
MULTIPLE OUTPUT MINIMIZATION



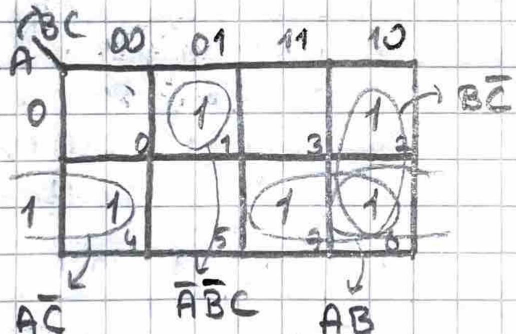
ÖRN $f_1 = \Sigma m(0,3,4,5,6)$

$f_2 = \Sigma m(1,2,4,6,7)$

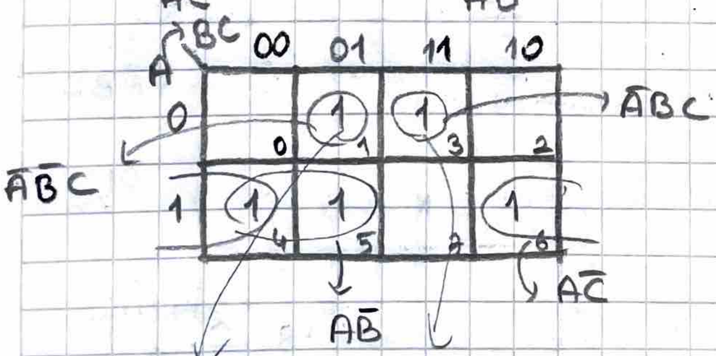
$f_3 = \Sigma m(1,3,4,5,6)$



$f_{1\,SOP} = \underbrace{A\bar{C}}_{\Sigma} + A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$



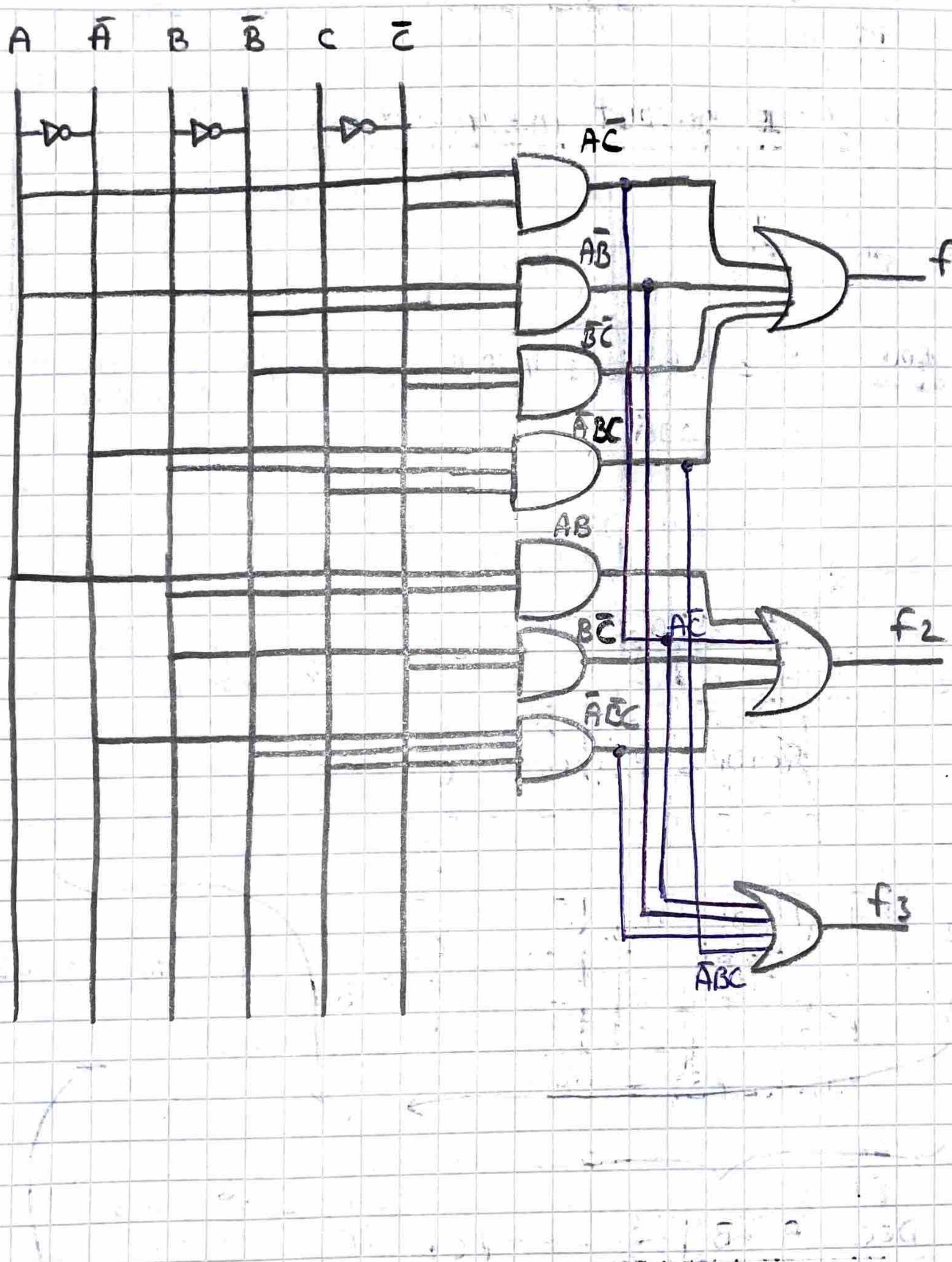
$f_{2\,SOP} = AB + \underbrace{A\bar{C}}_{\Sigma} + B\bar{C} + \boxed{\bar{A}\bar{B}C}$



$f_{3\,SOP} = A\bar{B} + \underbrace{A\bar{C}}_{\Sigma} + \boxed{\bar{A}\bar{B}C} + \bar{A}BC$

daha az maliyet için

⇒ Daha öncekilerde ayrı ayrı ald. için daha düşük maliyetli olur. Çünkü ekstra kapı kullanmayız! Bu yüzden tek tek aldık.


A
Ā
B
B̄
C
C̄
AC̄
ĀB̄
B̄C̄
ĀBC
AB
BC̄
AC̄
ĀB̄C
f1
f2
f3
ĀBC

ENTERED VARIABLE MAP REDUCTION

N : input variable

n : order of MAP (n < N)

N−n : free variables = Entered variables

2^{N-n} : Map key, Truth key

(Compressed map Cell Number) . (MAP key) ≑ (Code number of the first minterm of maxterm possible for that compressed mapp cell)

Önemli **ÖRN** $f(A,B,C) = \Sigma m(2,5,6,7)$ 2. mertebeden K-Map ile indirgeyiniz.

Dec	Dec	A (2^2)	B (2^1)	C (2^0)	f	
0	0	0	0	0	0	0
	1	0	0	1	0	
1	2	0	1	0	1	$\bar{C}$
	3	0	1	1	0	
2	4	1	0	0	0	C
	5	1	0	1	1	
3	6	1	1	0	1	1
	7	1	1	1	1	

Kaç tane yok edileceği (C'yi yok ettik)

N=3, n=2 → N−n = 3−2 = 1

$2^{N-n} = 2^1 = 2$

Kaç satırı teke indirgeyeceğiz

free variable : C

0 . 2 = 0 → 0,1
1 . 2 = 2 → 2,3
2 . 2 = 4 → 4,5
3 . 2 = 6 → 6,7

Hücre no

Ağırlığı düşük olanı seçeriz,,

A'nın ağırlığı : 2^2
B'nin ağırlığı : 2^1
C'nin ağırlığı : 2^0

Bu yüzden C

Dec	A	B	f
0	0	0	0
1	0	1	$\bar{C}$
2	1	0	C
3	1	1	1

Sıkıştırılmış haritanın doğruluk tablosu

A \ BC	00	01	11	10
0	0_{0}	0_{1}	0_{3}	1_{2}
1	0_{4}	1_{5}	1_{7}	1_{6}

f

C	
0	0
1	0

cell 0 → 0

C	
0	1
1	0

cell 1 → $\bar{C}$

C	
0	0
1	1

cell 2 → C

⇓

A \ B	0	1
0	0_{0}	$\bar{C}_{1}$
1	C_{2}	1_{3}

⇒

A \ B	0	1
0	0_{0}	$\bar{C}_{1}$
1	C_{2}	$C+\bar{C}_{3}$

AC $B\bar{C}$

A \ B	0	1
0	C	C $\bar{C}$
1	C	1

$A+\bar{C}$ B+C

$$f_{SOP} = AC + B\bar{C} \qquad f_{POS} = (A+\bar{C})\cdot(B+C)$$

<u>ÖRN</u> $f(A,B,C,D) = \prod M(0,1,6,8,9,11,14,15)$ 2. mertebeden K-map kullanınız.

AB \ CD	00	01	11	10
00	0_{0}	0_{1}	1_{3}	1_{2}
01	1_{4}	1_{5}	1_{7}	0_{6}
11	1_{12}	1_{13}	0_{15}	0_{14}
10	0_{8}	0_{9}	0_{11}	1_{10}

$N = 4$, $n = 2$ ⟩ $N-n = 2$ → free variable C, D

$2^{N-n} = 2^2 = 4$ → $16/4 = 4$

$0 \cdot 2^2 = 0$ → 0, 1, 2, 3
$1 \cdot 2^2 = 4$ → 4, 5, 6, 7
$2 \cdot 2^2 = 8$ → 8, 9, 10, 11
$3 \cdot 2^2 = 12$ → 12, 13, 14, 15

C \ D	0	1
0	0_{0}	0_{1}
1	1_{2}	1_{3}

/0: $C\bar{D}$ CD

C \ D	0	1
0	1_{4}	1_{5}
1	0_{6}	1_{7}

/1: $CD + \bar{C}\bar{D} + \bar{C}D$

C \ D	0	1
0	0_{8}	0_{9}
1	1_{10}	0_{11}

/2: $C\bar{D}$

C \ D	0	1
0	1_{12}	1_{13}
1	0_{14}	0_{15}

/3: $\bar{C}\bar{D} + \bar{C}D$

0 → $C\bar{D} + CD$ 1 → $CD + \bar{C}\bar{D} + \bar{C}D$ 2 → $C\bar{D}$ 3 → $\bar{C}\bar{D} + \bar{C}D$

Devamı

A \ B	0	1
0	$C\bar{D} + CD$	$CD + \bar{C}D + \bar{C}\bar{D}$
1	$C\bar{D}$	$\bar{C}D + \bar{C}\bar{D}$

$\bar{A}CD$

$\bar{C}D + \bar{C}\bar{D}$

$B(\bar{C}D + \bar{C}\bar{D}) = B\bar{C}\underbrace{(D+\bar{D})}_{1} = B\bar{C}$

$\bar{B}C\bar{D}$

$f_{SOP} = \bar{B}C\bar{D} + \bar{A}CD + B\bar{C}$

2.yol POS FORM

A \ B	0	1
0	$(C+D).(C+\bar{D})$	$(\bar{C}+D)$
1	$(C+D).(C+\bar{D})$ $(\bar{C}+\bar{D})$	$(\bar{C}+D)(\bar{C}+\bar{D})$

$\bar{B}+\bar{C}+D$

$\bar{A}+\bar{C}+\bar{D}$

$$B + (C+D).(C+\bar{D}) = B + C.C + C.D + C.\bar{D} + \cancel{D\bar{D}}^{\,0}$$

$$= B + C.C + C.\cancel{(D+\bar{D})}^{\,1}$$

$$= B + C\,\cancel{(C+C)}^{\,C} = B + C$$

$$f_{POS} = [B + (C+D).(C+\bar{D})].[\bar{B}+\bar{C}+D].[\bar{A}+\bar{C}+\bar{D}]$$

$$f_{POS} = (B+C).(\bar{B}+\bar{C}+D)(\bar{A}+\bar{C}+\bar{D})$$

NOR kapılarıyla gerçekleştirin derse bu yol

ÖRN 3 öğretmen 3 sorudan 40, 40, 20 puanlık bir sınav kağıdını 100 üzerinden değerlendiriyor. 1. ve 2. öğretmen 40 puanlık soruları 3. öğretmen ise 20 puanlık soruyu aşağıdaki şartlara göre değerlendiriyor

1. öğretmen → 0, 15, 25, 40 puan veriyor.
2. öğretmen → 0, 20, 30, 40, puan veriyor.
3. öğretmen → 0, 20 puan veriyor.

Öğrencinin başarı durumunu gösteren devreyi yalnızca NAND kapılarını kullanarak minimum maliyetle olacak şekilde gerçekleştirin.

Başarılı ≥ 50
4. mertebeden K-map ile

$n=5$ → L. C. → $m=1$

1. öğretmen

A_1	A_0	Puan
0	0	0
0	1	15
1	0	25
1	1	40

4 durum old. için (0, 15, 25, 40)

2. öğretmen

B_1	B_0	Puan
0	0	0
0	1	20
1	0	30
1	1	40

4 durum old. için (0, 20, 30, 40)

3. öğretmen

C	Puan
0	0
1	20

2 durum old. için (0, 20)

$N = 5$
$n = 4$

$N - n = 5 - 4 = 1$ $\quad 2^{N-n} = 2^1 = 2$

free variable yani C

free variable (serbest değişken)

1 → başarılı
0 → başarısız

başarılı ≥ 50

	Dec	A_1	A_0	B_1	B_0	C	f	
0	0	0	0	0	0	0	0	0
	1	0	0	0	0	1	0	
	2	0	0	0	1	0	0	
1	3	0	0	0	1	1	0	0
	4	0	0	1	0	0	0	
2	5	0	0	1	0	1	1	C → ≥50 (yeni başarılı)
	6	0	0	1	1	0	0	
3	7	0	0	1	1	1	1	C
	8	0	1	0	0	0	0	
4	9	0	1	0	0	1	0	0
	10	0	1	0	1	0	0	
5	11	0	1	0	1	1	1	C
	12	0	1	1	0	0	0	
6	13	0	1	1	0	1	1	C
	14	0	1	1	1	0	1	
7	15	0	1	1	1	1	1	$C+\overline{C}$ (yani 1)
	16	1	0	0	0	0	0	
8	17	1	0	0	0	1	0	0
	18	1	0	0	1	0	0	
9	19	1	0	0	1	1	1	C
	20	1	0	1	0	0	1	
10	21	1	0	1	0	1	1	$C+\overline{C}$
	22	1	0	1	1	0	1	
11	23	1	0	1	1	1	1	$C+\overline{C}$
	24	1	1	0	0	0	0	
12	25	1	1	0	0	1	1	C
	26	1	1	0	1	0	1	
13	27	1	1	0	1	1	1	$C+\overline{C}$
	28	1	1	1	0	0	1	
14	29	1	1	1	0	1	1	$C+\overline{C}$
	30	1	1	1	1	0	1	
15	31	1	1	1	1	1	1	$C+\overline{C}$

Karnaugh

A_1,A_0 \ B_1,B_0	00	01	11	10
00	0	0	C	C
01	0	C	$\bar{C}+C$	C
11	C	$C+\bar{C}$	$\bar{C}+C$	$C+\bar{C}$
10	0	C	$\bar{C}+C$	$C+\bar{C}$

$(C+\bar{C})\,A_0B_1B_0$
$= A_0B_1B_0$

C'leri aldık
A_0B_0C

B_1C

C'leri aldık
A_1A_0C

$A_1A_0B_0$

C'leri aldık
A_1B_0C

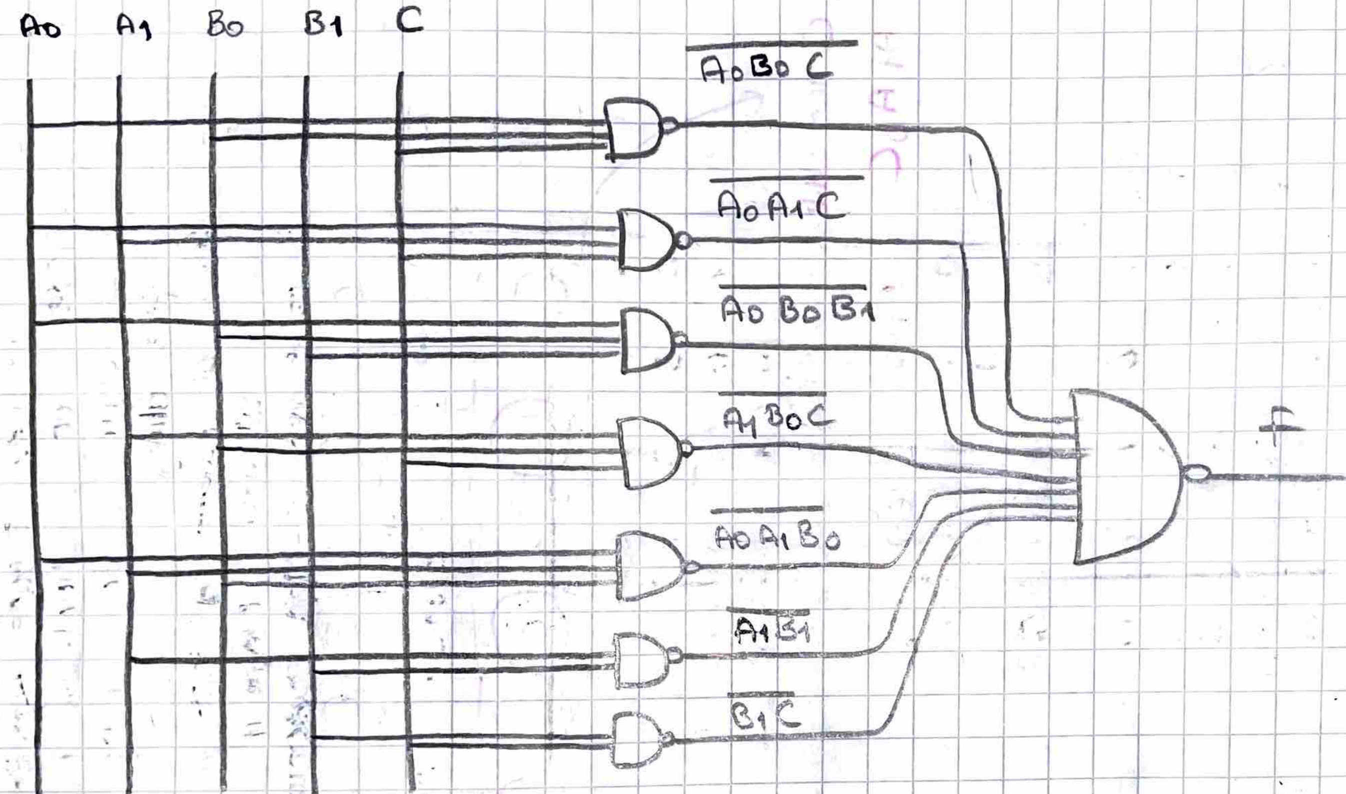
$\overbrace{(C+\bar{C})}^{1}(A_1B_1)$
$= A_1B_1$

$f = A_0B_0C + A_1A_0C + A_0B_1B_0 + A_1B_0C + A_1A_0B_0 + A_1B_1 + B_1C$

NAND kapılarıyla olması için 2 kez değilini alıyoruz,

$\bar{\bar{f}} = \overline{\overline{A_0B_0C} \cdot \overline{A_1A_0C} \cdot \overline{A_0B_1B_0} \cdot \overline{A_1B_0C} \cdot \overline{A_1A_0B_0} \cdot \overline{A_1B_1} \cdot \overline{B_1C}}$

= Nand kapısıyla çizim → sonraki sayfada


A0
A1
B0
B1
C
$\overline{A_0 B_0 C}$
$\overline{A_0 A_1 C}$
$\overline{A_0 B_0 B_1}$
$\overline{A_1 B_0 C}$
$\overline{A_0 A_1 B_0}$
$\overline{A_1 B_1}$
$\overline{B_1 C}$
F

ÖRN Bir hastanede 5 tane asansör vardır. 5. asansör yedek olarak kullanılıyor. 4 asansörden 2 veya daha fazla sayıda asansör meşgul olduğunda 5. asansör devreye alınıyor.
Böyle bir sisteme ait lojik devreyi yalnızca NOR kapıları kullanarak minimum maliyetle gerçekleştiriniz.
3. mertebeden K-Map kullanınız.

lojik 0'ları indirger

$n=4$ → L.C. → $m=1$ f

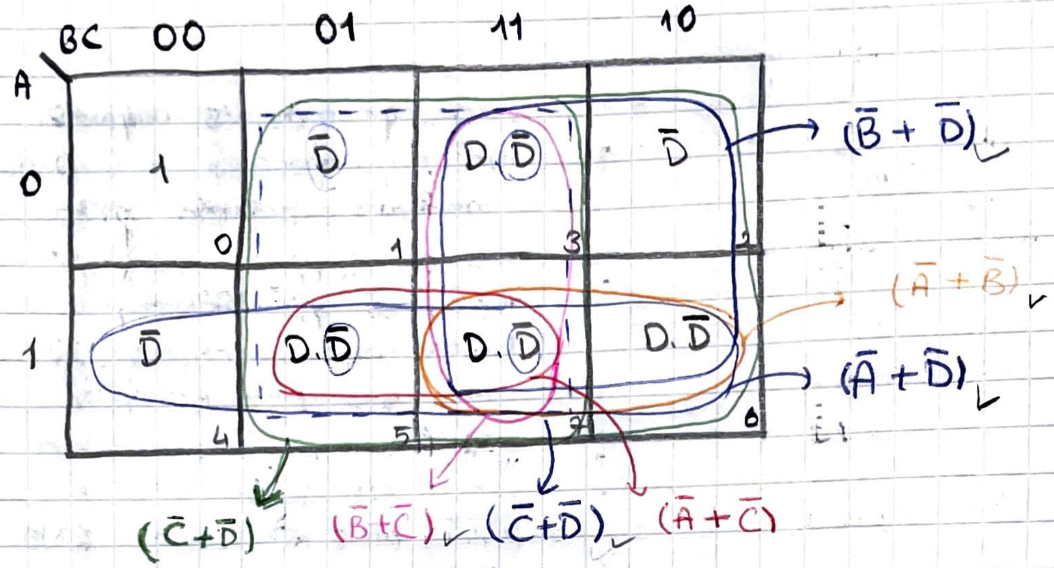
Çıkış
5. asansör | Çıkış 0'da aktif (NOR kapıları)

	Dec	A	B	C	D	f	
	0	0	0	0	0	1	1
0	1	0	0	0	1	1	
	2	0	0	1	0	1	
1	3	0	0	1	1	0	$\overline{D}$
	4	0	1	0	0	1	
2	5	0	1	0	1	0	$\overline{D}$
	6	0	1	1	0	0	
3	7	0	1	1	1	0	0
	8	1	0	0	0	1	
4	9	1	0	0	1	0	$\overline{D}$
	10	1	0	1	0	0	
5	11	1	0	1	1	0	$0 \to D.\overline{D}$
	12	1	1	0	0	0	
6	13	1	1	0	1	0	$0 \to D.\overline{D}$
	14	1	1	1	0	0	
7	15	1	1	1	1	0	$0 \to D.\overline{D}$

5. asansör devrede → 0

$N = 4$ $\qquad N-n = 4-3 = 1$ (free variable)
$n = 3$

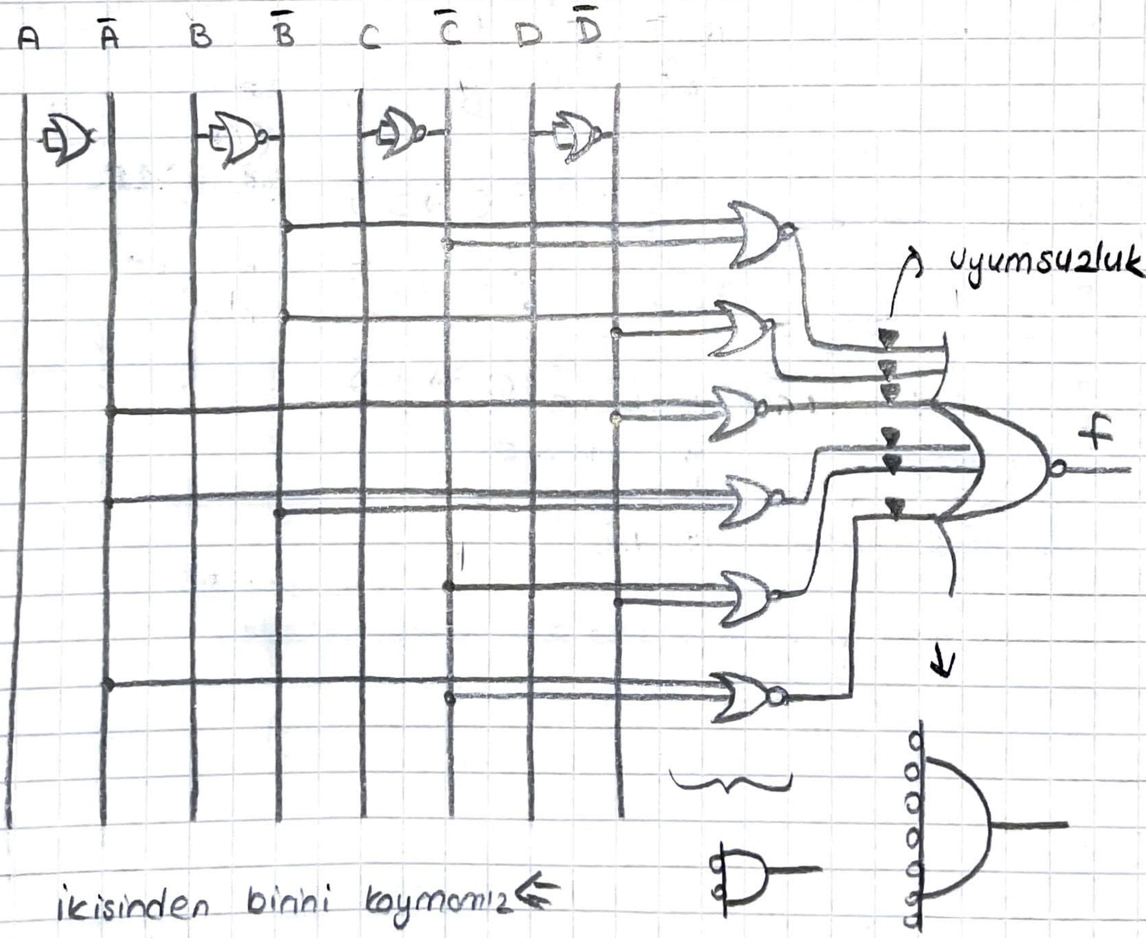
$2^{N-n} = 2^1 = 2$ (map key)



$$f_{POS} = (\bar{B}+\bar{C}).(\bar{B}+\bar{D}).(\bar{A}+\bar{D}).(\bar{A}+\bar{B}).(\bar{C}+\bar{D}).(\bar{A}+\bar{C})$$

$$\bar{\bar{f}} = \overline{\overline{(\bar{B}+\bar{C})} + \overline{(\bar{B}+\bar{D})} + \overline{(\bar{A}+\bar{D})} + \overline{(\bar{A}+\bar{B})} + \overline{(\bar{C}+\bar{D})} + \overline{(\bar{A}+\bar{C})}}$$

$$f_{POS} = \overline{x+y+z+w+m+N} \; \} \text{ NOR}$$



ikisinden birini koymamız gerek,,